

Introduction to Operating Systems - Assignment 2

Fall 2026

Questions and Answers

1) What are the system call names for getting the parent process ID, opening a file, closing a file, reading a file, printing to the console and sleeping?

- getppid()
- open()
- close()
- read()
- write()
- sleep()

2) What are the number of system calls for opening, closing and reading the file(s) (i.e. how many times each was called)?

- open() : 3 system calls
- close() : 3 system calls
- read() : 3 system calls

3) What are the number of system calls for printing to the screen? (count each individually. You may either use strace options to aid you in doing so, or you may use grep).

- 2 system calls

4) What was the value of the file descriptor of your read file (please review the lecture slides before asking what this means)?

- The value of the file descriptor is 3.

Source Code

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <fcntl.h>
4 #include <stdlib.h>
5 #include <time.h>
6
7 int main(int argc, char *argv[]) {
8
9     if(argc < 2) { // argv[0] is the program name and argv[v] should be the file name
10         printf("User forgot to provide a file name.\n");
11         return 1;
12     }
```

```

12     }
13
14     char ppid_msg[64];
15     int ppid_bytes = snprintf(ppid_msg, sizeof(ppid_msg), "Parent Process ID: %d\n", getppid
16         ());
17     write(STDOUT_FILENO, ppid_msg, ppid_bytes);
18
19     srand(time(NULL));
20     int random_sleep = rand()%10;
21     sleep(random_sleep);
22
23     // O_RDONLY means to open the file for reading only.
24     int file = open(argv[1], O_RDONLY);
25
26     if(file == -1) {
27         perror("Failed to open the file.\n");
28         return 1;
29     }
30
31     char buffer[1024];
32     size_t input_txt_bytes;
33
34     while((input_txt_bytes = read(file, buffer, sizeof(buffer))) > 0) {
35         write(STDOUT_FILENO, buffer, input_txt_bytes);
36     }
37
38     close(file);
39
40     return 0;
41 }

```

System Call Count (strace -c Log)

```
pomme@pomme-VMware-Virtual-Platform:~/Desktop/OS Homework/Assignment 2$ strace -c ./a.out
input.txt
Parent Process ID: 22675
echo "Hello World" > input.txt
echo "This is lab 2" > input.txt
% time      seconds      usecs/call      calls      errors      syscall
-----
0.00      0.000000          0          3          read
0.00      0.000000          0          2          write
0.00      0.000000          0          3          close
0.00      0.000000          0          2          fstat
0.00      0.000000          0          8          mmap
0.00      0.000000          0          3          mprotect
0.00      0.000000          0          1          munmap
0.00      0.000000          0          1          brk
0.00      0.000000          0          2          pread64
0.00      0.000000          0          1          1 access
0.00      0.000000          0          1          execve
0.00      0.000000          0          1          getppid
0.00      0.000000          0          1          arch_prctl
0.00      0.000000          0          1          set_tid_address
0.00      0.000000          0          1          clock_nanosleep
0.00      0.000000          0          3          openat
0.00      0.000000          0          1          set_robust_list
0.00      0.000000          0          1          prlimit64
0.00      0.000000          0          1          getrandom
0.00      0.000000          0          1          rseq
-----
100.00    0.000000          0          38          1 total
```

Detailed Strace Log

```

pomme@pomme-VMware-Virtual-Platform:~/Desktop/OS Homework/Assignment 2$ strace ./a.out input
.txt
execve("./a.out", ["/a.out", "input.txt"], 0x7fff48ca6468 /* 81 vars */) = 0
brk(NULL)                               = 0x576363539000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fb39d79c000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=74151, ...}) = 0
mmap(NULL, 74151, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fb39d789000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0e\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) =
840
fstat(3, {st_mode=S_IFREG|0755, st_size=2190608, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0e\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) =
840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fb39d400000
mmap(0x7fb39d428000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x28000) = 0x7fb39d428000
mmap(0x7fb39d5c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) =
0x7fb39d5c0000
mmap(0x7fb39d60e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0
x20e000) = 0x7fb39d60e000
mmap(0x7fb39d614000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1,
0) = 0x7fb39d614000
close(3)                                 = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fb39d786000
arch_prctl(ARCH_SET_FS, 0x7fb39d786740) = 0
set_tid_address(0x7fb39d786d68)         = 22420

```

```

set_robust_list(0x7fb39d786a20, 24) = 0
rseq(0x7fb39d786680, 0x21, 0, 0x53053053) = 0
mprotect(0x7fb39d60e000, 16384, PROT_READ) = 0
mprotect(0x57635e6df000, 4096, PROT_READ) = 0
mprotect(0x7fb39d7e1000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x87\x72\x59\x4d\x82\xa7\x01\x20", 8, GRND_NONBLOCK) = 8
munmap(0x7fb39d789000, 74151) = 0
getppid() = 22418
write(1, "Parent Process ID: 22418\n", 25Parent Process ID: 22418
) = 25
clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=9, tv_nsec=0}, {tv_sec=6, tv_nsec=565055792}) = ?
ERESTART_RESTARTBLOCK (Interrupted by signal)
--- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
restart_syscall(<... resuming interrupted clock_nanosleep ...>) = ? ERESTART_RESTARTBLOCK (
Interrupted by signal)
--- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
restart_syscall(<... resuming interrupted clock_nanosleep ...>) = ? ERESTART_RESTARTBLOCK (
Interrupted by signal)
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Interrupted by signal)
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restart_syscall(<... resuming interrupted clock_nanosleep ...>) = ? ERESTART_RESTARTBLOCK (
Interrupted by signal)
--- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
restart_syscall(<... resuming interrupted clock_nanosleep ...>) = ? ERESTART_RESTARTBLOCK (
Interrupted by signal)
--- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
restart_syscall(<... resuming interrupted clock_nanosleep ...>) = ? ERESTART_RESTARTBLOCK (
Interrupted by signal)
--- SIGWINCH {si_signo=SIGWINCH, si_code=SI_KERNEL} ---
restart_syscall(<... resuming interrupted clock_nanosleep ...>) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY) = 3
read(3, "echo \"Hello World\" > input.txt\n"..., 1024) = 64
write(1, "echo \"Hello World\" > input.txt\n"..., 64echo "Hello World" > input.txt
echo "This is lab 2" > input.txt
) = 64
read(3, "", 1024) = 0
close(3) = 0
exit_group(0) = ?
+++ exited with 0 +++

```

Screenshots

Program Execution

```
• pomme@pomme-VMware-Virtual-Platform:~/Desktop/OS Homework/Assignment 2$ gcc lab2.c
• pomme@pomme-VMware-Virtual-Platform:~/Desktop/OS Homework/Assignment 2$ ./a.out input.txt
Parent Process ID: 16405
echo "Hello World" > input.txt
echo "This is lab 2" > input.txt
```

Figure 1: Program execution showing parent process ID and file contents

Strace System Call Counts

```
• pomme@pomme-VMware-Virtual-Platform:~/Desktop/OS Homework/Assignment 2$ strace -c ./a.out input.txt
Parent Process ID: 22675
echo "Hello World" > input.txt
echo "This is lab 2" > input.txt
% time      seconds  usecs/call   calls    errors syscall
-----
 0.00      0.000000         0         3         0    read
 0.00      0.000000         0         2         0    write
 0.00      0.000000         0         3         0   close
 0.00      0.000000         0         2         0   fstat
 0.00      0.000000         0         8         0   mmap
 0.00      0.000000         0         3         0 mprotect
 0.00      0.000000         0         1         0 munmap
 0.00      0.000000         0         1         0   brk
 0.00      0.000000         0         2         0 pread64
 0.00      0.000000         0         1         1  access
 0.00      0.000000         0         1         0  execve
 0.00      0.000000         0         1         0  getppid
 0.00      0.000000         0         1         0 arch_prctl
 0.00      0.000000         0         1         0 set_tid_address
 0.00      0.000000         0         1         0 clock_nanosleep
 0.00      0.000000         0         3         0  openat
 0.00      0.000000         0         1         0 set_robust_list
 0.00      0.000000         0         1         0 prlimit64
 0.00      0.000000         0         1         0  getrandom
 0.00      0.000000         0         1         0   rseq
-----
100.00     0.000000         0        38         1 total
```

Figure 2: Output of the strace -c command counting system calls

Detailed Strace Output

[illegible]

Figure 3: Detailed execution trace using strace